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Année scolaire 2023/2024

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« Characterization of Coordinated Behavior in Online Social Media »

Effectué du 06/05/2024 au 27/07/2024 à João Monlevade



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Documento assinado digitalmente
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Data: 01/10/2024 07:42:38-0300
Verifique em <https://validar.iti.gov.br>

Acknowledgements

I would like to express my deep gratitude to everyone who contributed to the success of this internship.

First of all, I would like to warmly thank my internship tutor, Carlos Henrique Gomes FERREIRA, for his supervision, his valuable advice, and his availability throughout this experience. Your expertise and support allowed me to acquire new skills and better understand the realities of the profession. Thank you for guiding me with patience and giving me the opportunity to participate in enriching projects.

I would also like to thank UFOP for allowing me to use its premises and resources. Access to these infrastructures was essential for the smooth running of my work and greatly facilitated the accomplishment of my missions. I am grateful to the entire administrative and technical team who were always available to meet my needs.

Finally, I thank all the members of the UFOP team for their warm welcome, their collaboration, and the team spirit they demonstrated. This experience was extremely enriching on a professional and personal level, and I leave with many memories and learnings that will be valuable to me in the rest of my career.

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Glossary

API: Application Programming Interface – A set of definitions and protocols that allow different software applications to communicate with each other. It enables the integration of services, data, and functionalities across different platforms or systems.

AI: Artificial Intelligence – Refers to machine learning models and systems capable of performing tasks that typically require human intelligence.

ANSES: ANSES: Agence Nationale de Sécurité Sanitaire de l'Alimentation, de l'Environnement et du Travail – The French Agency for Food, Environmental and Occupational Health & Safety.

BDD: Base de Données (Database) – A structured collection of data stored and accessed electronically.

CDF: Cumulative Distribution Function – A statistical tool used to describe the probability distribution of a variable.

e-cig: Electronic Cigarette – A device that vaporizes a liquid solution, usually containing nicotine, for inhalation.

F1-score: A measure of a model's accuracy, combining precision and recall into a single score.

ISEN: Institut Supérieur d'Électronique et du Numérique – A higher education institution specializing in electronics and digital technologies.

UFOP: Federal University of Ouro Preto – A Brazilian university involved in this research collaboration.

V3 API: YouTube Data API (Version 3) – An API used to retrieve video and comment data from YouTube.

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Introduction

In our era, video content platforms have become essential for the dissemination and sharing of information, particularly because their reach is global and their real-time accessibility eliminates geographic and language barriers, allowing users to consume , create and share information in various formats. Whether through educational tutorials, documentaries or live news broadcasts, these platforms cater to a wide range of interests and needs, improving information retention and persuasion through visual engagement. As information circulates more and more quickly, video content platforms have established themselves as essential tools for education, awareness and societal influence.

Among the wide variety of topics discussed online, electronic cigarettes, also known as e-cigarettes, have sparked a global debate. These electronic devices deliver nicotine through inhaled vapor rather than traditional smoke, presenting both potential benefits and concerns. Proponents suggest that e-cigarettes offer a less harmful alternative to smoking, aiding tobacco cessation. However, critics warn of health risks, particularly long-term effects and their appeal to young people, which could lead to nicotine addiction. In France, this debate is particularly marked, with government and health authorities putting in place regulations to govern the use of e-cigarettes, including advertising restrictions, age limits and bans on smoking in public. France's cautious approach aims to balance public health benefits with protection of vulnerable populations.

A crucial question therefore arises in this debate: does the lack of complete and accessible information on electronic cigarettes not increase the population's uncertainty and doubt regarding their effects? Indeed, this scarcity of clear data leaves the public perplexed, struggling to understand the true benefits and risks of using e-cigarettes. As a result, the legitimacy of e-cigarettes over traditional cigarettes remains unclear and questionable for many. The lack of well-documented studies and research complicates public health discourse and decision-making, as policymakers and healthcare professionals find it difficult to provide definitive advice.

To fill this gap, creating a dataset focused on the French context and using the YouTube platform as a primary source of data could prove invaluable. YouTube, with its large user base and diverse content, offers a unique window into public opinion and discourse around e-cigarettes in France. By systematically collecting and analyzing video content, comments and discussions, researchers can obtain information on French consumers' perception of e-cigarettes, their concerns and their understanding of the associated risks and benefits. Such a dataset would provide a nuanced view of public attitudes, reveal trends over time and highlight demographic differences, contributing to a more informed understanding of e-cigarette use in France. This resource would strengthen the capacity of academics, policymakers, and health professionals to address the complexities of e-cigarette use, ultimately supporting more effective public health strategies and decision-making.

Company presentation

The Federal University of Ouro Preto (UFOP) is a Brazilian higher education establishment composed of 3 campuses, the headquarters is in Ouro Preto and the other two are located in the towns of Mariana and João Monelavde, all three located in the state of Minas Gerais. Its creation dates back to August 21, 1969, it resulted from the merger of two other higher education establishments.

UFOP is located in a city listed as a UNESCO World Heritage Site, which gives it a unique setting, between culture and education.

Regarding training, UFOP offers a wide range of first, second and third cycle programs, covering many disciplines such as engineering, pharmacy, social sciences, arts and letters, natural sciences or still healthy. It is also well known for its mining and metallurgical engineering programs, due to its heritage with the Minas school.

In addition, UFOP stands out in the field of research through its contributions in various scientific fields. The university is home to several cutting-edge research centers and actively encourages innovation and international collaboration. In particular, it participates in research projects in partnership with Brazilian and foreign institutions, which strengthens its position on the global academic scene.

UFOP also offers a student life very rich in cultural, social and sporting activities. Students benefit from an environment conducive to their intellectual and personal development, with numerous opportunities for community projects, student clubs or cultural events. The university also organizes festivals and events to celebrate the historical and cultural heritage of the region, thus strengthening the identity of the school.

UFOP is therefore a key player in the educational and socio-economic development of the Minas Gerais region and Brazil in general. Thanks to its numerous quality programs, its innovative research and its commitment to the community, UFOP contributes significantly to the training of competent and committed professionals, as well as to the production of knowledge that benefits all of society.

Context

This project is part of the collaboration between the Higher Institute of Electronics and Digital (ISEN) and the Federal University of Ouro Preto (UFOP). It is part of a major research initiative carried out under the direction of Professor Carlos Henrique Gomes Ferreira, which aims to explore the dynamics of discussions around electronic cigarettes in several countries, with a particular focus on France. The objective is to understand how electronic cigarettes are perceived, debated and understood in different cultural and regulatory contexts.

A first part of the work has already been carried out for Brazil, in Portuguese, as well as in English. The analysis of these speeches revealed interesting specificities depending on the linguistic and cultural contexts. Continuing this approach, we have chosen to concentrate our efforts on the French part in order to complete this comparative study. The French part of the research focuses on the collection and analysis of data from video content platforms, in particular YouTube, which offers a rich source of information on perceptions and public debates around electronic cigarettes.

This project is also part of the perspective of a scientific publication. The objective is to produce an article that will contribute to the scientific literature on perceptions and debates relating to electronic cigarettes around the world. This publication aims to share our results with the international scientific community, while informing policy makers and health professionals of current dynamics in this area.

During this internship, I worked closely with another student, adopting a collaborative approach at every stage of the project: from collecting data via YouTube to analyzing and interpreting it. This collaboration was crucial to comprehensively address the complex discourse surrounding electronic cigarettes, particularly in France. Together, we developed a robust dataset, based on videos, comments, and online discussions, which will be used not only for current research, but also for future studies. This work will contribute to a better understanding of the impact of electronic cigarettes in different national contexts.

In summary, this project represents a significant contribution to research on electronic cigarettes, taking into account the cultural and regulatory specificities of each country studied. It also paves the way for scientific publications that could influence public policies and public health strategies internationally.

Technical objective

I-Data collection

The technical objective of this project is to constitute a relevant and representative dataset of the perceptions and attitudes of the French public towards electronic cigarettes, by exploiting comments generated by users on the YouTube platform. To achieve this goal, it is essential to collect rich qualitative data, reflecting the diversity of opinions, personal experiences and discussions surrounding e-cigarettes, using a systematic and rigorous method.

The approach adopted is based on the use of a YouTube V3 API coupled with a personalized crawler, which makes it possible to collect comments related to electronic cigarettes in a targeted manner. The first step is to identify and extract relevant comments using a list of carefully selected keywords. These keywords should cover a wide range of terms associated with e-cigarettes, including popular brands, technical terms relating to the devices, as well as common expressions and synonyms related to nicotine consumption and health debates.

The data collection period extends from 2018 to the end of 2023, a period long enough to capture changes in public discourse and identify emerging trends. The objective is to create an exhaustive and reliable database, which can serve as a basis for subsequent studies on perceptions of electronic cigarettes in France. This database will not only paint a general picture of public opinions, but also reveal recurring concerns and nuances in discussions, such as risk perceptions, personal experiences, and attitudes towards regulations in force.

Thus, the ultimate goal is to have a robust dataset, capable of providing valuable insights for in-depth analyzes and contributing to a better understanding of how e-cigarettes are perceived and debated in the French context.

Technical objective

II-Data cleaning

The technical objective of the data cleaning phase is to ensure that the dataset collected from YouTube comments on e-cigarettes is of high quality, relevant, and specifically adapted to the analysis of perceptions in France. To achieve this, several stages of rigorous filtering and cleaning are implemented to ensure that the final data accurately reflects opinions and discussions relevant to our research topic.

The first step is to detect and remove duplicate comments, in order to avoid any distortion of the results which could skew the analysis by amplifying the frequency of certain points of view. Next, it is crucial to filter off-topic comments and videos, i.e. those not directly related to e-cigarettes, using a combination of prohibited keywords to eliminate irrelevant content, such as than advertisements or discussions about other products.

Once the video database has been cleaned, the next step is to ensure that all retained comments are in French. This is essential so that our artificial intelligence can work effectively on content exclusively in French. To achieve this goal, two AI models are used in tandem to filter comments: first the Roberta model, then a language detection model. These two models successively analyze each comment to identify those which are in French. Only comments validated by both AIs are included in the final dataset, reducing the risk of linguistic errors and ensuring that our analysis focuses only on French content.

In short, the objective is to produce a cleaned, precise dataset, and specifically adapted to the analysis of perceptions of electronic cigarettes in France. This cleaning process, with its multiple verification steps, aims to create a solid and reliable database that will serve as a foundation for in-depth analyzes and future research.

Technical objective

III-Deep learning model

After cleaning the database and filtering the comments to keep only those in French, the next step consists of training an artificial intelligence model capable of determining whether a comment is favorable or unfavorable to the use of electronic cigarettes. This step involves several technical and methodological challenges, in particular the creation of a labeling file which will serve as a training basis for our model.

Creation of the Labeling File

To use the AI model effectively, it is essential to have a manually labeled data set. This labeling file is crucial to ensure the quality of model training. Around 4,000 comments are randomly selected from the database, and each is classified into one of three categories:

1. **Favorable Comments:** Those who express support or a positive opinion regarding electronic cigarettes.
2. **Unfavorable Comments:** Those who express opposition or criticism towards electronic cigarettes.
3. **Neutral/Undetermined Comments:** Those which do not make it possible to clearly determine a position for or against the use of electronic cigarettes, or which are ambiguous.

In order to maximize the quality of the marking and, therefore, the accuracy of the training, this process was carried out by three different people. Each commenter annotated the same sample of 400 comments, ensuring that the annotations are consistent and reliable.

To determine the final set of labels and minimize discrepancies between annotators, we used the Fleiss Kappa method. This statistical index indicates the degree of agreement between multiple raters on a data set, taking into account the likelihood that annotations are produced by chance. This method makes it possible to quantify the reliability of the annotations and ensure that the labeling file reflects a consensus between the annotators.

AI Model Training

Once the labeling file has been created, it serves as the basis for training the AI model. This is based on advanced deep learning and neural network techniques. The main challenge here is ensuring that the model correctly learns to distinguish the subtle nuances between different types of comments, while avoiding biases that could skew the results.

Technical objective

III-Deep learning model

Training a model of this type involves several subtleties and challenges:

1. **Data Bias:** If the labeling file contains an imbalance in the number of favorable or unfavorable comments, the model risks becoming biased, favoring one polarity over the other. It is therefore crucial to maintain a balance in the training data set.
2. **Linguistic Complexity:** French comments may contain idiomatic expressions, sarcasm, or ambiguous wording that may be difficult to correctly interpret by an AI model. Training must therefore be sophisticated enough to capture these subtleties.
3. **Overfitting:** Another risk is that of overfitting, where the model becomes overfit to the training data and loses its ability to generalize to new feedback. This requires the use of regularization and cross-validation techniques to evaluate the model's performance on unseen data.
4. **Evolution of Language:** The language used in comments may evolve over time, with new expressions or terms appearing. The model must be regularly updated to incorporate these developments and remain relevant.

Once the model is trained and validated, it will be used to automatically classify the remaining comments in the database, providing an overview of public opinions on e-cigarettes in France. This process will generate valuable insights and contribute to a better understanding of attitudes and perceptions towards electronic cigarettes, while paving the way for more in-depth analyses.

Technical objective

IV- Data analysis

In this section, we will define the technical objectives of analyzing data from YouTube comments on e-cigarettes. The goal is to deeply explore the cleaned dataset using various statistical and visual methods to draw conclusions about major trends, perceptions and themes.

Characterization of Engagement and Content Production Metrics

We will start by analyzing engagement and content production metrics using cumulative distribution functions (CDF). These CDFs will allow us to visualize the distribution of user interactions and contributions. By examining how comments are distributed in terms of popularity and activity, we will be able to gain insight into user engagement with e-cigarette-related content.

Heat Maps of Associated Terms

We'll then use heat maps to analyze terms associated with brands, flavors, and other relevant themes. Heatmaps will provide a visualization of the most frequently encountered terms in comments and help identify key associations between these terms. This visual approach will make it easier to understand the dominant themes and relationships between different aspects of e-cigarette discussions.

Heat Maps for Terms Relating to Sales, Regulations and Health Agencies

Another set of heat maps will focus on terms relating to sales, regulations and health agencies. These maps will highlight variations and trends associated with these specific aspects of the discussions. They will allow you to visualize how topics related to sales, regulation and health agencies are discussed in the comments and how their importance evolves over time.

AI Performance Analysis Using F1-Score

To evaluate the performance of our artificial intelligence model in classifying comments, we will use the F1-score. The F1-score is a measure of a model's accuracy on a dataset, calculated as the harmonic mean of the precision and recall. It provides a balanced metric for evaluating both false positives and false negatives, making it particularly useful for imbalanced datasets.

Technical objective

IV- Data analysis

During analysis, the F1-score will help us evaluate the accuracy of our model by identifying terms that appear significantly more or less frequently than expected. This will allow us to determine how well our model detects relevant terms and adjusts its predictions based on the data distribution.

Analysis of Comment Types and Word Clouds

Finally, we will look at the most common types of comments through word clouds. Word clouds will provide a visual representation of the most commonly used terms in comments, with an emphasis on topics related to disease, government, and other related themes. This analysis will identify the main concerns expressed in the comments and understand the contexts in which these concerns are raised.

In summary, this section aims to provide a comprehensive and detailed analysis of the dataset using statistical and visual methods. The results obtained will make it possible to identify trends, identify key themes and better understand the discussions around electronic cigarettes in France

Technical development

I-Data collection

Data Coverage Period: 2020 to 2023

The data collection process covers a period from 2020 to 2023. The aim is to capture the evolution of discussions and perceptions around e-cigarettes on YouTube over the years. Data for the years 2022 and 2023 have already been collected, providing a solid basis for analyzing recent trends.

Identification of Required Channels and Videos

We targeted a total of 150,384 YouTube channels, which represents a broad representation of content creators dealing with e-cigarettes. From these channels, we identified 502,940 relevant videos for analysis. These videos were selected based on their relevance to the defined search terms, ensuring that only content directly related to e-cigarettes was included.

Extraction of Comments

The core of our data collection is based on extracting comments from selected videos. In total, 14,175,433 comments were extracted. These comments constitute the raw material for our analysis, as they directly reflect users' opinions, discussions and attitudes about e-cigarettes.

Using the Official YouTube API

To ensure the accuracy and reliability of our data collection, we used the official YouTube API. This API allows structured and programmatic access to YouTube data, facilitating the extraction of necessary information while respecting the constraints and limitations imposed by the platform.

Using the official API has several advantages:

- **Structured Data Access:** Allows information to be retrieved with precise formatting, thus facilitating subsequent processing.
- **Compliance with YouTube Policies:** Ensures that our data collection complies with YouTube's terms of service, thereby avoiding sanctions or restrictions.
- **Advanced Filtering Possibility:** Offers filtering options based on metadata such as publication date, number of views, or interactions, which allows you to refine the selection of relevant videos and comments.

Technical development

I-Data collection

Definition of Search Strings

Defining search strings is a crucial step in our collection process. It determines the key terms used to identify videos and comments relating to e-cigarettes. These strings have been developed to cover a wide range of terms related to e-cigarettes, including popular brand names, technical terms relating to the devices, and phrases commonly associated with the health debate and regulation of e-cigarettes.

Examples of Search Strings:

"cigarette électronique"

"vape"

"nicotine"

"vaporisateur"

"sevrage tabagique"

"risques pour la santé"

These channels were integrated into queries sent via the YouTube API to identify relevant videos and comments. They help ensure that our data collection is comprehensive and captures the different dimensions of the debate around electronic cigarettes.

Summary of the Collection

The collection process resulted in an extensive and diverse database spanning four years of discussions on YouTube. Thanks to the use of the official YouTube API and a precise definition of search strings, we now have a valuable resource to analyze perceptions and attitudes towards e-cigarettes in France. This systematic and structured collection will serve as a solid foundation for the next steps of our project, including data cleaning, analysis, and training of artificial intelligence models.

Technical development

II-Data cleaning

Data cleaning is a crucial step to ensure the quality and relevance of our database, especially in the context of analyzing e-cigarette reviews. This process includes several sub-steps, each aimed at removing unwanted elements and refining the data to make it usable by our analysis models.

Duplicate Elimination and Data Preparation

The first step in our cleaning process is to remove duplicate videos and restrict the scan period. We start by identifying duplicates in our database based on the video ID.

```
video_df = videos_df.drop_duplicates(subset=['video_id'])  
video_df['published_at'] = video_df['published_at'].str.slice(0, 10)  
start_date = '2018-01-01'  
end_date = '2023-12-31'  
video_df = video_df[(video_df['published_at'] >= start_date) & (video_df['published_at'] <= end_date)]
```

Figure 1 : Duplicate elimination and data preparation

This step ensures that each video is unique and only videos published between 2018 and 2023 are considered.

Stopword Filtering and Keyword Search

Next, we filter video titles using a personalized list of stopwords in French and English, as well as relevant keywords to identify videos related to e-cigarettes.

```
stopwords_english = set(stopwords.words('english'))  
stopwords_french = set(stopwords.words('french'))  
custom_stopwords = set(["Acer", "acer", "ACER"])  
final_stopwords = stopwords_english.union(stopwords_french, custom_stopwords)
```

Figure 2 : Stopwords setup

Technical development

II-Data cleaning

```
keywords = [
    "cigarette", "cigarettes", "cigarette électronique", "cigarettes électroniques",
    "nicotine", "nicotines", "saveur", "saveurs", "mod", "mods", "résistance", "résistances",
    "vapoter", "vapote", "vapotes", "vapoteur", "vapoteurs", "vapoteuse", "vapoteuses", "jus",
    "réservoir", "réservoirs", "batterie", "batteries", "liquide", "liquides", "cartouche",
    "cartouches", "atomiseur", "atomiseurs", "ecig", "ecigs", "e-liquide", "e-liquides",
    "kit de démarrage", "kits de démarrage", "juul", "juuls", "smok", "geekvape", "voopoo",
    "aspire", "eleaf", "innokin", "vuse", "vaporesso", "freemax", "joyetech", "vape", "vapes",
    "e-cigarette", "e-cigarettes", "flavor", "flavors", "coil", "coils", "vaping", "vaper",
    "vapers", "vapeur", "vapeurs", "vapeuse", "vapeuses", "juice", "juices", "tank", "tanks",
    "battery", "batteries", "liquid", "liquids", "pod", "pods", "pen", "pens", "cloud", "clouds",
    "atomizer", "atomizers", "ejuice", "ejuices", "eliquid", "eliquids", "starter kit", "starter kits",
    "flavor chaser", "flavor chasers", "nic salt", "nic salts", "freebase nicotine",
    "freebase nicotines", "elfbar", "elfbars", "elf bar", "elf bars", "Ignite", "Ignites", "Uwell",
    "Uwells", "Caliburn", "Caliburns", "Argus", "Arguses", "DOTMOD", "DOTMODS", "GEEK VAPE",
    "GEEK VAPES", "Innokin", "Innokins", "Joyetech", "Joyetechs", "Justfog", "Justfogs",
    "taux de nicotine", "taux de nicotines", "tabac chauffé", "tabacs chauffés", "glycerine",
    "glycerines", "glycerol", "glycerols", "drip-tip", "drip-tips", "mesh", "meshes", "dl", "mtl",
    "dry-hit", "dry-hits", "throat hit", "throat hits", "vaporisateur", "vaporisateurs", "puff",
    "puffs", "cloud chaser", "cloud chasers", "flacon", "flacons", "fil résistif", "fils résistifs",
    "mod box", "mod boxes", "mod mécanique", "mods mécaniques", "rta", "rtas", "rda", "rdas",
    "rdta", "rdtas", "e-cig", "e-cigs", "vape pen", "vape pens", "vape mod", "vape mods", "dripper",
    "drippers", "clearomiseur", "clearomiseurs", "sub-ohm", "sub-ohms", "microcoil", "microcoils",
    "cotonnage", "cottonnages", "vapeur", "vapeurs", "e-vape", "e-vapes", "power vaping", "nasty",
    "oxva", "oxvas", "xim pro", "xim pros"
]

keywords = [word for word in keywords if word not in final_stopwords]
```

Figure 3 : List of keywords

Keywords are chosen to cover terms commonly associated with electronic cigarettes, excluding custom stopwords to ensure the relevance of the results.

Filtering Videos by Keywords and Excluding Irrelevant Titles

We identify relevant videos by filtering those whose titles contain at least one of the defined keywords. At the same time, we exclude videos whose titles contain irrelevant terms to avoid the inclusion of off-topic content.

```
videos_with_keyword = video_df[video_df['title'].str.contains('|'.join(keywords), case=False, na=False)]

exclude_words = [
    "podcast", "music", "time", "lyric", "Andaman", "Nicobar", "island", "Alan Walker",
    "Music Video", "type beat", "Lucid Dream", "Juice WRLD", "WRLD", "rugby", "Mobile Legend",
    "drink", "asmr", "ft", "feat", "Weed", "Food", "vlog", "cbd", "acer", "Joe Rogan", "alan walker",
    "music", "street food", "recette", "Official music", "day", "big", "king", "walker", "easy", "lyric"
]

# Create a regex pattern for the words to exclude
exclude_pattern = r'\b(?:' + '|'.join(re.escape(word) for word in exclude_words) + r')\b'

# Filter out videos with titles containing any of the exclude words
videos_with_keyword = videos_with_keyword[~videos_with_keyword['title'].str.contains(exclude_pattern, case=False, na=False)]
```

Figure 4 : Filtered videos by keywords and excluding irrelevant titles

Technical development

II-Data cleaning

This filtering ensures that only relevant videos are retained, excluding irrelevant content identified by the excluded keywords.

Validation and Visualization by Word Cloud

Finally, to check the quality of the cleaned data, we use word clouds to visualize the most frequent terms in the video titles selected. This allows us to confirm that the titles are indeed related to electronic cigarettes and that extraneous topics have been eliminated.

```

videos_with_keywords = video_df[video_df['title'].str.contains(' '.join(keywords), case=False, na=False)]
videos_without_keywords = video_df[~video_df['title'].str.contains(' '.join(keywords), case=False, na=False)]
videos_with_excluded_words = videos_with_keywords[videos_with_keywords['title'].str.contains(' '.join(exclude_words), case=False, na=False)]
filtered_videos = videos_with_keywords[~videos_with_keywords['title'].str.contains(' '.join(exclude_words), case=False, na=False)]
filtered_titles = filtered_videos['title']
excluded_titles = pd.concat([videos_without_keywords, videos_with_excluded_words], axis=0)['title']
def clean_titles(titles):
    return titles.apply(lambda x: ' '.join([word for word in x.split() if word.lower() not in final_stopwords]))

cleaned_excluded_titles = clean_titles(excluded_titles)
cleaned_filtered_titles = clean_titles(filtered_titles)

def plot_word_cloud(titles, file_name):
    wordcloud = WordCloud(width=1920, height=1080, background_color='white').generate(' '.join(titles))
    plt.figure(figsize=(10, 5))
    plt.imshow(wordcloud, interpolation='bilinear')
    plt.axis('off')
    plt.savefig(file_name, bbox_inches='tight')
    plt.show()

plot_word_cloud(cleaned_excluded_titles, 'figs/cleaned_excluded_titles.png')
plot_word_cloud(cleaned_filtered_titles, 'figs/cleaned_filtered_titles.png')

```

Figure 5 : Creation of the word clouds

II-Data cleaning



Figure 6 : word cloud of non-cleaned titles



Figure 7 : word cloud of cleaned titles

This filtering ensures that only relevant videos are retained, excluding irrelevant content identified by the excluded keywords.

Technical development

II-Data cleaning

Filtering Comments in French

Once the videos are filtered, we focus on the comments to only keep those written in French. This process uses two cascading AI models: Roberta and a language detection model.

```
# Load the model and tokenizer for language detection
model_ckpt = "paplucu/xlm-roberta-base-language-detection"
tokenizer = AutoTokenizer.from_pretrained(model_ckpt)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = AutoModelForSequenceClassification.from_pretrained(model_ckpt).to(device)

# Load previous state if exists
dict_comment_language = load_state()

# We will only filter comments belonging to the pre-selected videos and within the target range date
df_comments = pd.read_csv("raw_data/comments_info_corrected.csv", on_bad_lines='warn', encoding='utf-8', sep=',', lineterminator='\n')
videos_with_keyword = pd.read_csv("cleaned_data/filtered_videos_with_keywords_substring.csv")
df_comments['comment'] = df_comments['comment'].astype(str)
df_comments['published_at'] = pd.to_datetime(df_comments['published_at'])
start_date = '2018-01-01'
end_date = '2023-12-31'
df_comments = df_comments[(df_comments['published_at'] >= start_date) & (df_comments['published_at'] <= end_date)]
df_comments = df_comments[df_comments['video_id'].isin(videos_with_keyword['video_id'])]
dict_comments_to_detect_language = dict(zip(df_comments['comment_id'].astype(str), df_comments['comment']))
del videos_with_keyword

# Ensure dict_comment_language keys are strings
dict_comment_language = {str(k): v for k, v in dict_comment_language.items()}

# Identify unprocessed comment IDs
unprocessed_comment_ids = [comment_id for comment_id in dict_comments_to_detect_language if comment_id not in dict_comment_language]

# Print debug information
print(f"Total comments to process: {len(dict_comments_to_detect_language)}")
print(f"Total processed comments: {len(dict_comment_language)}")
print(f"Total unprocessed comments: {len(unprocessed_comment_ids)}")
```

Figure 8 : Roberta model

```
def detect_language(text):
    try:
        return detect(text)
    except:
        return 'unidentified' # Retorna isso se não conseguir detectar

df_comments = pd.read_csv("cleaned_data/comments_info_language_roBERTa_substring.csv", on_bad_lines='skip', encoding='utf-8', sep=',', lineterminator='\n')
df_comments['published_at'] = pd.to_datetime(df_comments['published_at'])
df_comments = df_comments.dropna(subset=['published_at'])
tqdm.pandas(desc="Detecting languages")
df_comments.loc[df_comments['language'] != 'fr', 'language_langdetect'] = df_comments[df_comments['language'] != 'fr']['comment'].progress_apply(detect_language)
df_comments.to_csv("cleaned_data/final_comments_all_language_vape.csv", index=False)
df_comments_fr = df_comments[(df_comments['language'] == 'fr') | (df_comments['language_langdetect'] == 'fr')]
print("Total of comments belonging to filtered videos and in french", len(df_comments_fr))
df_comments_fr.to_csv("cleaned_data/final_comments_fr_vape_substring.csv", index=False)
```

Figure 9 : Language detection model

The use of two successive models helps reduce errors and ensures that only comments in French are retained.

Technical development

III-Deep learning model

The process of developing a deep learning model to detect opinions on electronic cigarette use begins with a crucial labeling step. This step consists of creating a set of annotated data, essential for training the AI to recognize whether a comment is in favor, against, or neutral in relation to the use of electronic cigarettes.

Creation of the Annotated Dataset

To guarantee the quality of the annotation, we randomly selected a sample of around 4000 comments. Each comment is then manually classified into one of the following three categories:

Favorable: The comment clearly expresses support or a favorable opinion for the use of electronic cigarettes.

Contary: The comment expresses an unfavorable or critical opinion regarding electronic cigarettes.

Indifferent: The comment is neutral or does not make it possible to clearly determine a position for or against.

In addition, we also had videos which, despite the filtering previously carried out, were out of context or not in French. For these videos we used Remove video (RV) if the video was not related to electronic cigarettes and Remove Comment (RC) if the comments were in a language other than French.

To maximize the accuracy of the labeling, three different people annotated the same sample of 4000 comments. This approach provides a more robust overview and minimizes individual bias

Technical development

III-Deep learning model

	A	B	C
1	Title	Comment	Label
2	ON MANGE BLEU VS ROUGE PENDANT 24H ! (couleur préférée)	Attention avec la chèvre il faut pas la prendre comme ça ça peut lui faire mal même si vous faites doucement	RV
3	Pop Smoke - Demeanor feat. Dua Lipa (Official Video)	Je suis choqué de ce clip ! On y voit un enfant observer une partouze, un gangbang, c'est honteux, ça prône la pédophilie et le patriarcat matriarcat, domination et luxure ! Réveillez vous ! Ce sont des petites personnes aux pouvoirs qui font des choses atroces, qui influencent ce monde qu'on ne veut plus. DUALIPA n'est qu'un objet de ce monde, une petite fille manipuler pour faire vendre de la dope aux grands publics qui avalent car on la engraine dans le capitalisme et la culture de l'apparence. Regardez regardez bien et renseignez vous ! Ce clip est le reflet d'une part de notre monde et de ce qu'on vend aux jeunes, des faux rêves ! De la violence, de la comparaison et d'une hiérarchie qui accroît le jugement ect.... Vous voulez vraiment que vos enfants trouvent ça normal ! Stop !	RV
4	GEEKVAPE AEGIS POD (PH)	same question	RC
5	Décorticage : pourquoi la cigarette électronique jetable est une aberration écologique	WE... mais c'est trop à la mode!! un fumeur de gauchiste. C'est super intéressant toute cette électronique, je n'y connais absolument rien, à mon grand désarroi pourtant je bricole et touche à pas mal de choses, mais je n'ai jamais eu l'occasion ni le temps de me pencher et de m'intéresser sérieusement à ce domaine. J'ai quand même réussi à faire un tourne-broche maison à partir d'un moteur d'essuie-glace et d'un chargeur de batterie 😊	I
6	Ma première cigarette électronique 🤔🤔🤔🤔	Tu peux le trouver sur ce discount à 35€ désolé je n'est pas le lien	I
7	LA VAPE TUE ?! (Retour sur les dernières polémiques...)	Bordel mais les maniaques sont vraiment idiots il n'y a pas eu UNE personne qui est morte à cause de ça	F
8	DÉCOUVERTE 100% RESPECTUEUSE (non) DU DLC POKÉMON MASQUE TURQUOISE 🇹🇷	Leur potentiel était exceptionnel, comme BDSP, malheureusement ils ont fait de mauvais choix	RV
9	Si je voulais vapoter dans le monde	C'est vrai que c'est un problème grave aux États-Unis vous imaginez 6 décès potentiellement liés au vapotage? Mais c'est un problème très grave!!!! (Non je déconne c'est de l'ironie, ils devraient plutôt interdire les armes à feu et les clopes ces imbéciles)	F
10	OUVERTURE PUFF (je viens ouvrir ma chaîne doucement les critiques)	Oui d'accord Corentin	I
11	La vape chez les buralistes, mauvaise idée ?	Il a totalement raison ma mère et tabagiste et mon père barman et ma mère m'a dit que si elle vend des produits de la vape c'est pour faire des bénéfices	C
12	Cbd la pineapple express de bouddhisme+ grosse promo	Tiens toi prêt à être traduisible devant un tribunal 😡😡😡😡. Tu vas raquer tes grands morts. Tu fais moins le malin la 😡😡😡. (On vous surveille).	I
13	l'intérêt des drippers RDA	de rien	I
14	SHADOW OF THE COLOSSUS (PS4) - 15 - Argus	où est ce qu'on conseille des jeux sympas à MV ?	RV
15	Pop Smoke - Merci Beaucoup (Visualizer)	En vrai le son sans se mentir il pue la merde..	RV

Figure 10 : Exemple of labeling

Technical development

III-Deep learning model

Using the Fleiss Kappa Index

To determine the final set of labels, we used the Fleiss Kappa index method. This index makes it possible to measure the agreement between several annotators beyond simple chance, which is crucial for evaluating the consistency and reliability of the labeling.

- A high Fleiss Kappa index (near 1) indicates excellent agreement between annotators, while an index close to 0 would indicate poor agreement.
- The results of this agreement then allow us to decide whether a comment should be included in the final training dataset, based on consensus among the annotators.

```
def calculate_fleiss_kappa(list1, list2, list3):  
    # Combine the lists into a single list of lists  
    combined_list = list(zip(list1, list2, list3))  
  
    # Create a set of unique labels  
    labels = list(set(list1 + list2 + list3))  
  
    # Create an empty matrix for the ratings  
    n = len(combined_list)  
    k = len(labels)  
    ratings_matrix = np.zeros((n, k))  
  
    # Fill the ratings matrix  
    for i, ratings in enumerate(combined_list):  
        for rating in ratings:  
            j = labels.index(rating)  
            ratings_matrix[i][j] += 1  
  
    # Calculate the Fleiss' Kappa score  
    kappa_score = fleiss_kappa(ratings_matrix)  
  
    return kappa_score
```

Figure 11 : Calculation of Kappa Fleiss Index

Technical development

III-Deep learning model

Linguistic Challenges and Biases

When labeling, several language-related challenges can introduce bias. For example :

- Ambiguous expressions: Some terms or expressions can be interpreted differently depending on the context or perspective of the annotator.
- Sarcasm or irony: These figures of speech are difficult to detect and can be misinterpreted during annotation.
- Cultural differences: Cultural perceptions around electronic cigarettes may vary, thus influencing the classification of comments.

These challenges require careful attention during labeling to ensure that model training truly reflects the nuances of the language used.

Training the Deep Learning Model

With the labeled dataset, the deep learning model can be trained. The provided Python file uses a multilingual BERT model to classify comments.

```
def prepare_data(file_path, train_size, validation_size, test_size):
    df = pd.read_csv(file_path, usecols=['comment', 'label'])
    df['comment'] = df['comment'].astype(str)
    df['comment'] = df['comment'].apply(lambda x: x.replace('\n', ' ').replace('\t', ' ').strip())

    # Mapping the labels C, I, F to 0, 1, 2
    label_mapping = {'C': 0, 'I': 1, 'F': 2}
    df['label'] = df['label'].map(label_mapping)

    df = df.sample(frac=1).reset_index(drop=True)

    print(df.groupby("label").count())

    train_val_df, test_df = train_test_split(df, test_size=test_size, stratify=df['label'], random_state=SEED)
    train_df, val_df = train_test_split(train_val_df, test_size=validation_size/(train_size + validation_size), stratify=train_val_df['label'], random_state=SEED)

    train_dataset = Dataset.from_pandas(train_df)
    val_dataset = Dataset.from_pandas(val_df)
    test_dataset = Dataset.from_pandas(test_df)

    return train_dataset, val_dataset, test_dataset
```

Figure 12 : Preparation of data before deep learning

Technical development

III-Deep learning model

Training with Transformers:

The model is then trained with different combinations of hyperparameters to optimize accuracy.

```
# Configure the model and tokenizer
tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME)
model = AutoModelForSequenceClassification.from_pretrained(MODEL_NAME, num_labels=3, hidden_dropout_prob=DROP_OUT, attention_probs_dropout_prob=DROP_OUT)

training_args = TrainingArguments(
    output_dir='./results',
    num_train_epochs=int(EPOCH),
    per_device_train_batch_size=int(BATCH_SIZE),
    per_device_eval_batch_size=int(BATCH_SIZE),
    warmup_steps=500,
    weight_decay=float(WEIGHT_DECAY),
    logging_dir='./logs',
    learning_rate=float(LEARNING_RATE),
    evaluation_strategy="no", # No evaluation during final training
    save_strategy="epoch",
    save_total_limit=1,
)
```

Figure 13 : Configuration of the model and tokenizer

```
# Function to compute performance metrics
def compute_metrics(eval_pred):
    accuracy = load_metric("accuracy", trust_remote_code=True)
    precision = load_metric("precision", trust_remote_code=True)
    recall = load_metric("recall", trust_remote_code=True)
    f1 = load_metric("f1", trust_remote_code=True)
    logits, labels = eval_pred
    predictions = np.argmax(logits, axis=-1)

    precision_scores = precision.compute(predictions=predictions, references=labels, average=None)
    recall_scores = recall.compute(predictions=predictions, references=labels, average=None)
    f1_scores = f1.compute(predictions=predictions, references=labels, average=None)

    metrics_per_class = {"precision_class_{i}": p for i, p in enumerate(precision_scores["precision"])}
    metrics_per_class.update({"recall_class_{i}": r for i, r in enumerate(recall_scores["recall"])})
    metrics_per_class.update({"f1_class_{i}": f for i, f in enumerate(f1_scores["f1"])})

    return metrics_per_class
```

Figure 14 : Function to compute performance metrics

Technical development

III-Deep learning model

```
# Train the model
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset, # Evaluating on test set to keep validation during training
    compute_metrics=compute_metrics,
)

trainer.train()
```

Figure 15 : Train the model

Linguistic Bias Management

In a multilingual context, biases can emerge, especially when a model is trained mainly on French data. These biases can affect model performance, including its ability to correctly interpret comments with specific linguistic or cultural nuances. Using multilingual pre-trained models like BERT is crucial to minimizing these biases, but additional adjustments may be necessary, such as adding additional training data for French language dialects or variants.

Performance Evaluation:

To evaluate the performance of the model, we use standard metrics such as precision, recall and F1-score, calculated for each class (Favorable, Contary, Indifferent).

Technical development

III-Deep learning model

```
# Function to perform bootstrapping on the test set
def bootstrap_test(predictions, labels, n_iterations=BOOTSTRAP_ITERATIONS):
    rng = np.random.default_rng(BOOTSTRAP_SEED) # Initialize a random number generator
    idx = np.arange(labels.shape[0]) # Create an array of indices for the labels

    # Lists to store metric scores for each class
    precision_scores_class_0, recall_scores_class_0, f1_scores_class_0 = [], [], []
    precision_scores_class_1, recall_scores_class_1, f1_scores_class_1 = [], [], []
    precision_scores_class_2, recall_scores_class_2, f1_scores_class_2 = [], [], []

    # Perform bootstrapping
    for _ in tqdm(range(n_iterations), desc="Bootstrap Iterations"):
        pred_idx = rng.choice(idx, size=idx.shape[0], replace=True) # Resample the indices with replacement
        pred_boot = predictions[pred_idx] # Get the bootstrap sample of predictions
        labels_boot = labels[pred_idx] # Get the bootstrap sample of labels

        # Calculate precision, recall, and F1-score for class 0
        precision_class_0 = precision_score(labels_boot, pred_boot, labels=[0], average=None, zero_division=0)[0]
        recall_class_0 = recall_score(labels_boot, pred_boot, labels=[0], average=None, zero_division=0)[0]
        f1_class_0 = f1_score(labels_boot, pred_boot, labels=[0], average=None, zero_division=0)[0]

        # Calculate precision, recall, and F1-score for class 1
        precision_class_1 = precision_score(labels_boot, pred_boot, labels=[1], average=None, zero_division=0)[0]
        recall_class_1 = recall_score(labels_boot, pred_boot, labels=[1], average=None, zero_division=0)[0]
        f1_class_1 = f1_score(labels_boot, pred_boot, labels=[1], average=None, zero_division=0)[0]
```

Figure 16: Performing bootstrapping on the test set

```
# Calculate precision, recall, and F1-score for class 2
precision_class_2 = precision_score(labels_boot, pred_boot, labels=[2], average=None, zero_division=0)[0]
recall_class_2 = recall_score(labels_boot, pred_boot, labels=[2], average=None, zero_division=0)[0]
f1_class_2 = f1_score(labels_boot, pred_boot, labels=[2], average=None, zero_division=0)[0]

# Store the scores for each iteration
precision_scores_class_0.append(precision_class_0)
recall_scores_class_0.append(recall_class_0)
f1_scores_class_0.append(f1_class_0)

precision_scores_class_1.append(precision_class_1)
recall_scores_class_1.append(recall_class_1)
f1_scores_class_1.append(f1_class_1)

precision_scores_class_2.append(precision_class_2)
recall_scores_class_2.append(recall_class_2)
f1_scores_class_2.append(f1_class_2)
```

Figure 17: Show the metrics for each class

Technical development

III-Deep learning model

Mean, Standard Deviation, and Confidence Intervals of Metrics for each class:

	mean	std	lower_95_ci	upper_95_ci
precision_class_0	0.000000	0.000000	0.000000	0.000000
recall_class_0	0.000000	0.000000	0.000000	0.000000
f1_class_0	0.000000	0.000000	0.000000	0.000000
precision_class_1	0.959720	0.014694	0.928562	0.988097
recall_class_1	0.828027	0.026103	0.772705	0.875664
f1_class_1	0.888762	0.016700	0.852031	0.918210
precision_class_2	0.686692	0.040870	0.603297	0.765629
recall_class_2	0.946795	0.023077	0.898990	0.988239
f1_class_2	0.795278	0.029509	0.735039	0.849791

Figure 18: Metrics results for each class

Here we obtain a score for precision, recall and f1 score of 0 for our class 0, this is explained by the lack of sufficient "Contary" type data in our database not allowing us to carry out interesting analyzes .

For class 1:

Accuracy: The average is 0.959720, which shows that of the examples the model identified as class 1, about 96% were correct. The confidence interval [0.928562, 0.988097] indicates relatively small variation.

Recall: The average is 0.828027, which means that the model correctly identified about 88% of the examples in class 1. The standard deviation (0.026103) shows fairly good stability, but there is a slight uncertainty with the ' interval [0.772705, 0.875664].

F1-Score: At 0.888762 on average, the F1-score shows a good balance between precision and recall for class 1. The confidence interval [0.852031, 0.918210] indicates robust performance.

For class 2:

Accuracy: At 0.686692 on average, the accuracy is lower for this class, meaning the model had more difficulty classifying this class well. The confidence interval [0.603297, 0.765629] shows a larger uncertainty compared to class 1.

Recall: At 0.946795 on average, the model is much more efficient in identifying examples of class 2 (nearly 95%). The confidence interval [0.898990, 0.988239] shows a smaller variation here.

F1-Score: The F1 score is 0.795278, with some variability, as shown by the standard deviation of 0.029509. The interval [0.735039, 0.849791] indicates that this is the class for which the model could benefit from improvements.

Technical development

IV- Data analysis

Characterization of Engagement and Content Production Metrics using CDF

From the data provided on user engagement metrics (like the number of likes and comments per video), we generated a cumulative distribution function (CDF) to explore the distribution of user engagement with -Vis content related to electronic cigarettes. The code below generates a CDF graph of likes or comments.

```
favorable_df = filtered_df[filtered_df['label_predicted'] == 2]

# Group by author and count the number of comments for each author
favorable_users = favorable_df.groupby('author').size().reset_index(name='num_comments')

# Plot the CDF for the number of comments a channel receives
plt.figure(figsize=(6, 4))
sns.ecdfplot(data=favorable_users, x='num_comments', color='black')
plt.xlabel('# Comments (x)')
plt.ylabel('CDF')
plt.xscale('log') # Set the x-axis to logarithmic scale
plt.grid(True, which="both", ls="--")
plt.title('CDF of Comments by Favorable Users')
plt.show()
plt.savefig('CDF_comments_favorable_users.png')
```

Figure 19: Creation of the CDF of comments by favorable users

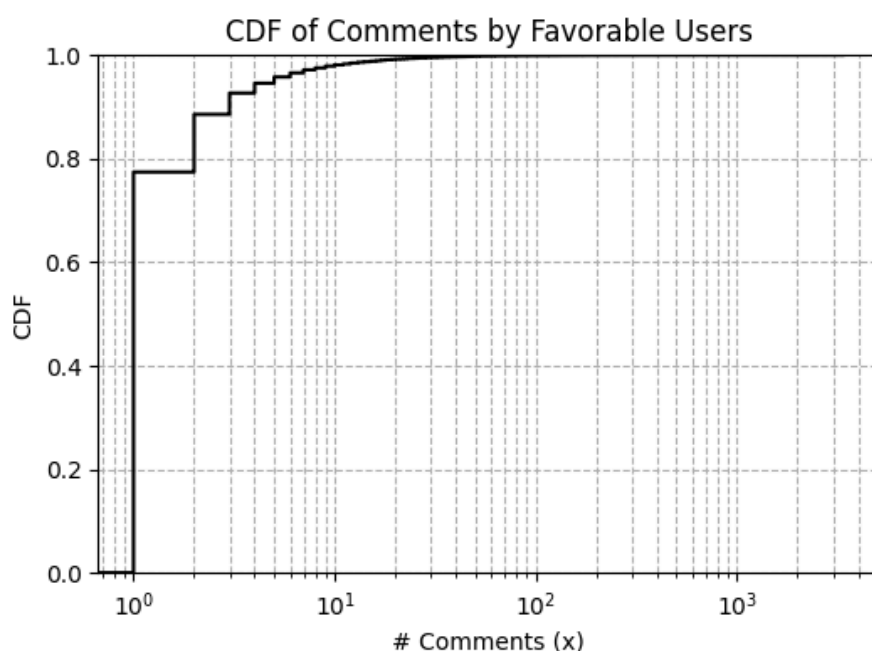


Figure 20: CDF of comments by favorable users

Technical development

IV- Data analysis

The results obtained from the CDF show that the majority of videos on electronic cigarettes obtain relatively few likes. The curve is tilted at the beginning, meaning that many videos receive a low number of likes (e.g., less than 100), while a small percentage of videos get a disproportionate number of likes.

This indicates that the distribution of likes is highly uneven, likely due to the fact that a few very popular videos dominate in terms of engagement, while most videos have limited interactions. This type of distribution is common on social media platforms, where a minority of content captures most of users' attention.

Heat Maps of Associated Terms

The script below extracts terms associated with brands and flavors mentioned in user comments on e-cigarette videos, then calculates a correlation matrix. This is visualized in the form of a heat map, making it possible to identify the relationships between the different brands and flavors.

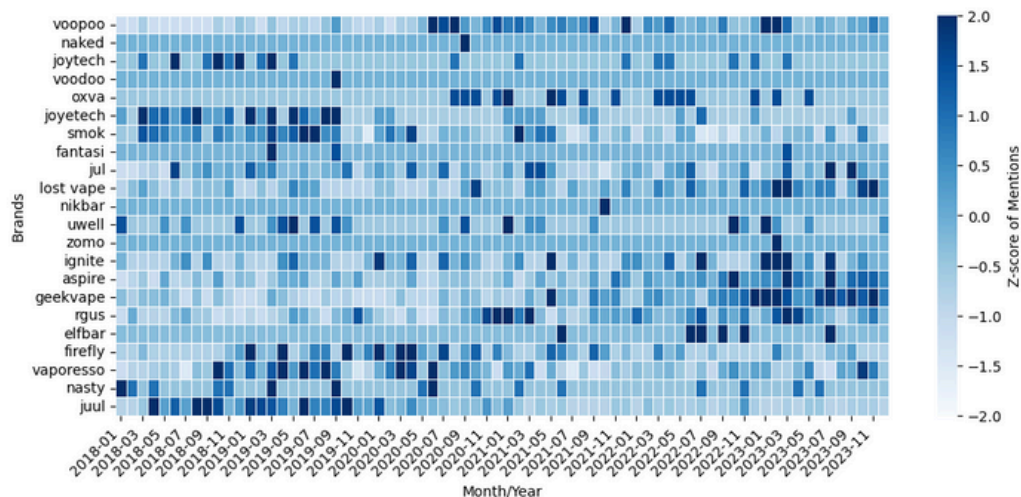


Figure 21: Heatmap of Brands

Technical development

IV- Data analysis

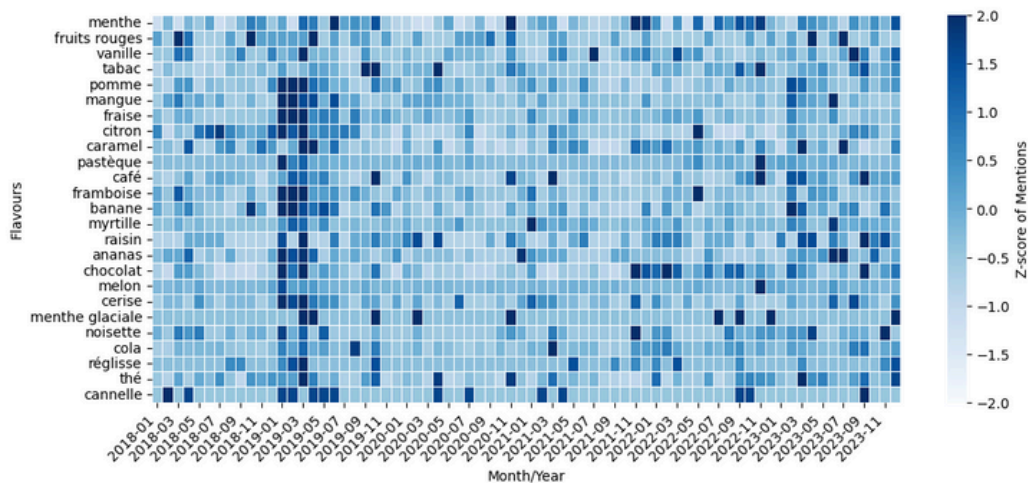


Figure 22: Heatmap of flavours

Heat Maps for Terms Relating to Sales, Regulations, and Health Agencies

In this section, heat maps were generated to analyze terms relating to sales, regulations and health agencies. Terms like “taxation,” “prohibition,” and “public health” are often mentioned in connection with debates around government regulations and concerns.

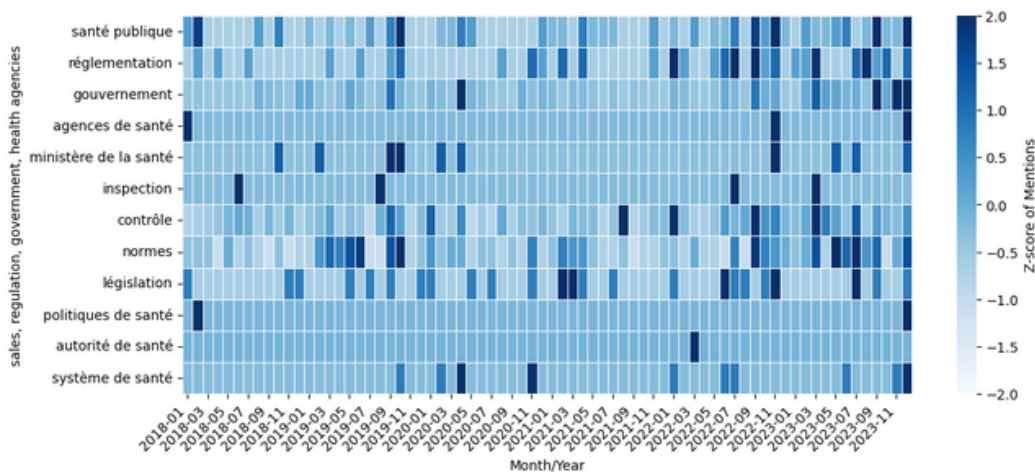


Figure 23: Heatmap of sales, regulations, government, health and agencies

Technical development

IV- Data analysis

Analysis of Comment Types and Word Clouds

We generated a word cloud from the comments to identify the most frequent terms and understand users' main concerns regarding e-cigarettes.



Figure 24: Word clouds of favorable comments



Figure 25: Word clouds of indifferent comments

Technical development

IV- Data analysis

We can observe that the words which appear most often in comments favorable to the use of electronic cigarettes will be terms such as “liquide” or even “vape” or “resistance” which are terms associated with the use of electronic cigarettes. world of electronic cigarettes, moreover, we can observe other terms like “hello” or “good” which are rather positive terms.

On the side of comments classified as indifferent, we observe terms like “video” or even “Juste” or “monde” which represent more random words without really having a positive or negative aspect, so we cannot really draw a trend of this.

Project management

This project, resulting from the collaboration between the Higher Institute of Electronics and Digital (ISEN) and the Federal University of Ouro Preto (UFOP), is part of an international research framework led by Professor Carlos Henrique Gomes Ferreira. The objective is to study the discursive dynamics around electronic cigarettes in several cultural and regulatory contexts, focusing particularly on France. The project is part of broader comparative work which already includes analyzes carried out in Brazil (in Portuguese) and in English-speaking countries.

The project was rigorously structured through different phases defined from the start of the process. A Gantt chart was created to visualize the key steps, including data collection, data analysis and reporting of results. Each step had specific deadlines, with a clear distribution of responsibilities among team members.

One of the central aspects of this project was the collaboration with another student. This partnership enabled an efficient distribution of tasks, ranging from the collection of data from YouTube to its analysis. Collaboration was essential to comprehensively address the complex discourse around electronic cigarettes, particularly in the French context. Team work facilitated the exchange of ideas and strengthened the quality of the analyzes carried out, while guaranteeing methodological consistency with previous studies carried out in Brazil and in English-speaking countries.

Under the supervision of Professor Carlos, the project took place mainly remotely, due to university strikes in Brazil. This constraint required rigorous remote organization, with the use of online communication tools to guarantee coordination and monitoring of the project. Despite these obstacles, the work proceeded efficiently thanks to regular meetings and open communication channels with the supervisor.

Although the project was part of a larger research initiative, it was strongly framed, with specific expectations and a methodological framework defined from the start. This structure, although limiting in terms of freedom of action, made it possible to be part of a dynamic consistent with research carried out in other cultural and linguistic contexts. The objective was to produce comparable results, which guided the methodological choices throughout the project.

The project will result in the creation of a scientific article intended for publication. This article will contribute to the existing literature on perceptions and debates around electronic cigarettes around the world, taking into account the cultural and regulatory specificities of each country. Additionally, this project paves the way for future research by creating a robust dataset that can be reused for other comparative studies.

Conclusion

Objective and Context

This report describes a research project aimed at characterizing coordinated behaviors related to electronic cigarettes on social networks, with particular emphasis on the French context. The main objective was to build a comprehensive dataset from YouTube comments and videos to reflect public perceptions, debates and attitudes towards e-cigarettes in France.

Data Collection

The data collection phase resulted in the creation of a large dataset . These videos were selected using specific keywords related to e-cigarettes, covering the period from 2018 to 2023. This helped capture the evolution of public discourse over time, with a particular focus on emerging trends and recent debates.

The use of the YouTube API guaranteed structured and filtered data extraction, respecting the constraints of the platform. This methodology provided a comprehensive and relevant body of data for analysis.

Data Cleaning

Data cleaning was a crucial step to ensure the quality and relevance of the final dataset. Duplicates, irrelevant content and non-French comments were systematically removed using two AI models: the Roberta model and a language detection model, working in tandem. This rigorous process ensured that the final dataset contained exclusively French-language content relating to e-cigarettes.

The cleaned dataset now provides a solid basis for subsequent analyzes and future research into public perceptions of e-cigarettes in France.

Development of a Deep Learning Model

After data cleaning, a deep learning model was developed to automatically classify comments into three categories: favorable, unfavorable or neutral towards the use of e-cigarettes.

Model development required the creation of a manually labeled dataset. Around 4,000 comments were randomly selected and categorized by three different annotators. This process was validated using Fleiss' Kappa method to ensure agreement between annotators, ensuring the reliability of the dataset for model training.

Conclusion

The model was then trained using advanced neural network techniques, taking into account linguistic nuances such as idiomatic expressions and sarcasm, common in French discussions about e-cigarettes. Special measures have been taken to avoid biases, such as imbalances between favorable and unfavorable comments, which could distort the performance of the model.

Data Analysis and Visualizations

Once the deep learning model was trained, it was applied to the cleaned dataset to automatically classify millions of comments. This analysis generated statistical indicators and visualizations revealing key trends in public discussions about e-cigarettes.

Cumulative Distribution Functions (CDFs): These functions were generated to explore the distribution of user engagement (e.g. likes and comments per video). The results showed a very uneven distribution, where a small number of videos generated the majority of interactions, a typical phenomenon on social media platforms where a minority of content captures the majority of audience attention.

Heat Maps: Heat maps were used to identify terms most commonly associated with e-cigarette brands, flavors, regulations, and health agencies. This visual representation helped identify dominant themes and associations within the discussions.

Word Clouds: Finally, word clouds were created to visualize the most frequent terms used in comments, highlighting key public concerns such as health risks and regulatory debates.

These analyzes provided a comprehensive overview of the French public's perceptions and attitudes toward e-cigarettes, revealing recurring themes and concerns expressed in online discussions.

Future Outlook

The work carried out within the framework of this project opens several perspectives for future research and technical improvements:

Expansion of International Comparative Studies: The methodology developed in this study can be applied to other linguistic and cultural contexts. By carrying out comparative studies with other countries, it would be possible to identify significant differences in public perceptions of electronic cigarettes depending on regulatory and cultural contexts. This comparative work could be valuable for developing public health policies adapted to each country.

Conclusion

Improved AI Models: Although the developed deep learning model performs well, there is room for improvement, particularly to better capture complex linguistic nuances, such as irony or region-specific expressions. By adding more diverse datasets or incorporating more sophisticated models, model accuracy could be improved to better capture the intricacies of online discussions.

Application to Other Subjects: The methodology used to analyze electronic cigarettes can be replicated for other societal or public health subjects, such as vaccination, environmental debates or political speeches. Understanding how coordinated behaviors emerge in these areas could provide crucial information for policymakers.

Continuous Enrichment of the Dataset: As the language evolves, it would be beneficial to regularly update the dataset to capture new terms, expressions and emerging trends. This would ensure the continued relevance of future analyses.

General Conclusion

In conclusion, this project established a systematic and rigorous approach to analyze public perceptions of electronic cigarettes, particularly in the French context. The methodology, from data collection to analysis via deep learning models, provides a solid foundation for future research and contributes to a better understanding of the public discourse on e-cigarettes. In addition to offering direct implications for policy makers and healthcare professionals, this project presents an analytical framework extensible to other topics, enriching research on coordinated online behaviors.

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Summary

Français :

Ce rapport présente les résultats d'un projet de recherche mené dans le cadre d'une collaboration entre l'Institut Supérieur d'Électronique et du Numérique (ISEN) et l'Université Fédérale d'Ouro Preto (UFOP). Sous la direction du Professeur Carlos Henrique Gomes Ferreira, le projet analyse les dynamiques de discussion autour des cigarettes électroniques dans plusieurs pays, avec une attention particulière pour la France. En concentrant l'analyse sur les commentaires et les vidéos disponibles sur YouTube, ce travail vise à comprendre les perceptions, les débats et les contextes culturels et réglementaires associés aux cigarettes électroniques. Les résultats fourniront des insights précieux pour les décideurs politiques et les professionnels de santé, tout en contribuant à la littérature scientifique sur ce sujet.

English:

This report presents the results of a research project conducted as part of a collaboration between the Higher Institute of Electronics and Digital (ISEN) and the Federal University of Ouro Preto (UFOP). Under the direction of Professor Carlos Henrique Gomes Ferreira, the project analyzes the dynamics of discussions surrounding electronic cigarettes across various countries, with a particular focus on France. By concentrating on comments and videos available on YouTube, this work aims to understand perceptions, debates, and the cultural and regulatory contexts associated with electronic cigarettes. The findings will provide valuable insights for policymakers and health professionals, while contributing to the scientific literature on this topic.